# Customer Shopping Behavior Analysis

Understanding purchasing patterns, demographics, and loyalty impact across 3,900 customers

Tarun Geebu

# Project Overview

### Python

Data cleaning and feature engineering engineering

### MySQL

Advanced SQL queries and analysis

### Power BI

Interactive dashboard visualization

Dataset: 3,900 customers × 19 data points

# Python Data Pipeline

Key Steps

- Data loading and inspection
- Missing values filled with median ratings
- Age binned into 4 demographic groups
- Redundant columns removed
- Exported to MySQL database

# SQL Analysis Highlights

**$157,890**

Male Revenue

vs. $75,191 Female

**4.9**

Top Rating

Shorts category

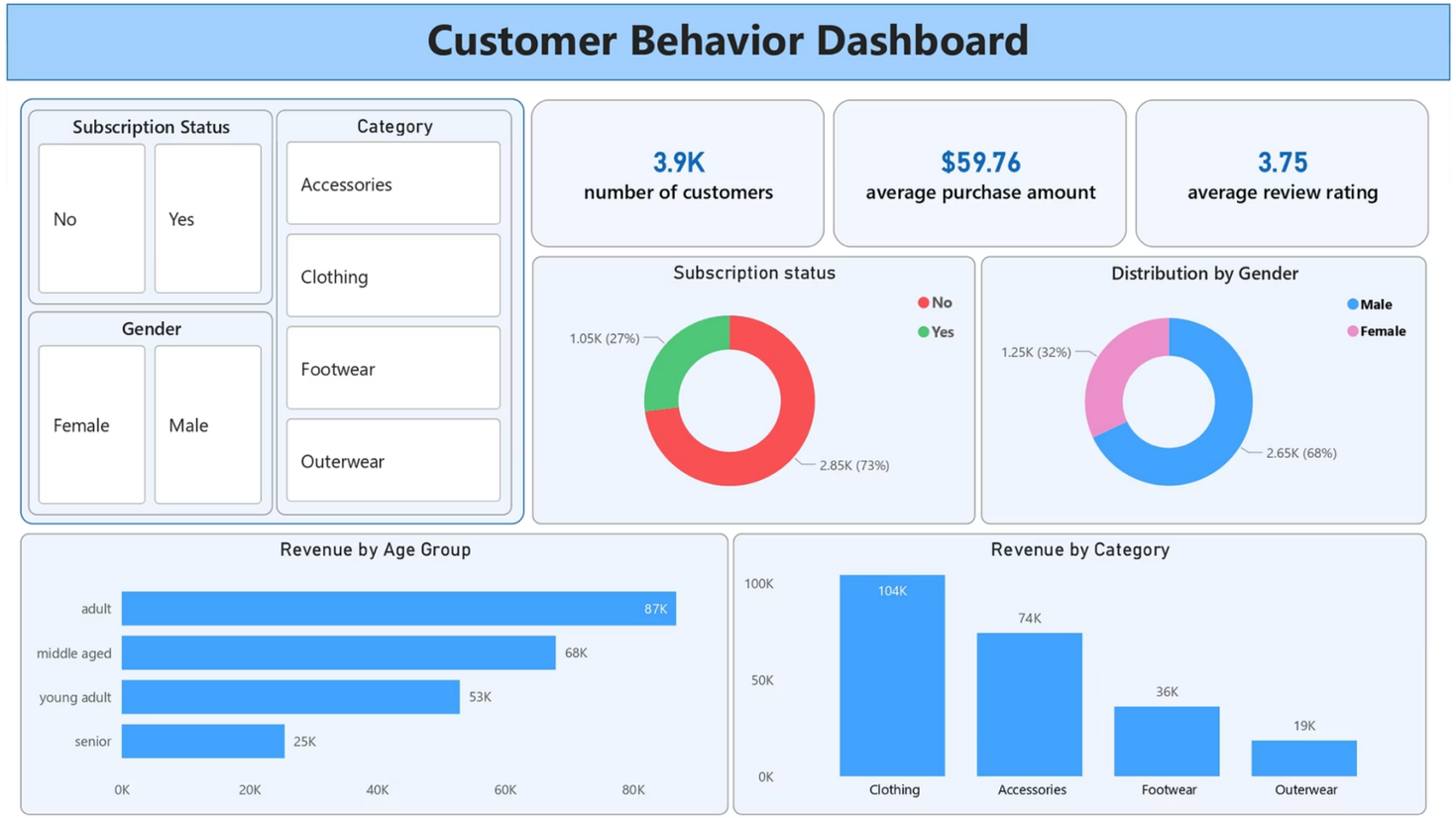
**23.5%**

Top Discount

Sweaters category

category

# Power BI Dashboard




### 3.9K Customers

Total customer base analyzed

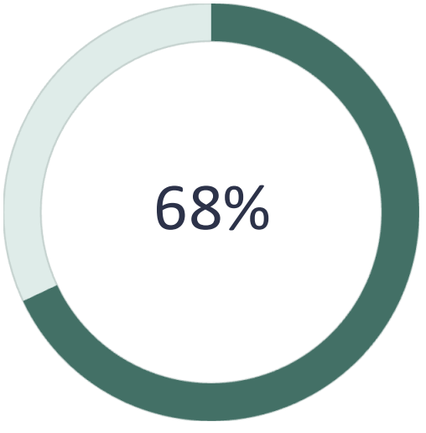
### $59.76

Average purchase amount

### 3.75 Rating

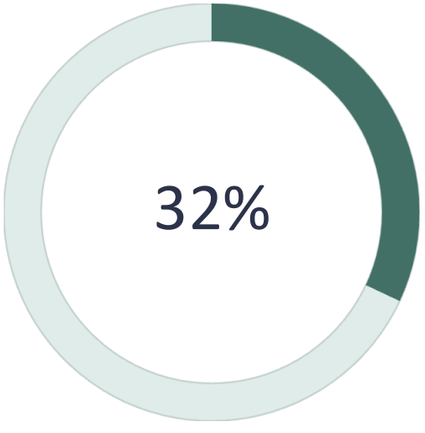
Average review score

# Customer Segmentation



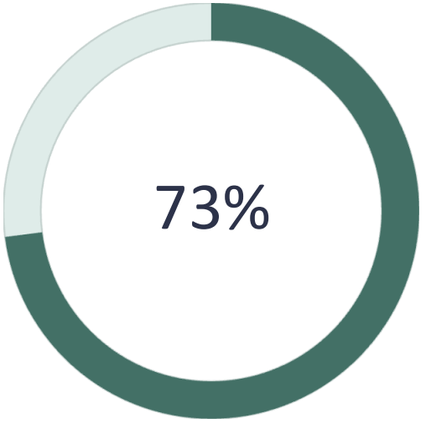

Male

2,650 customers



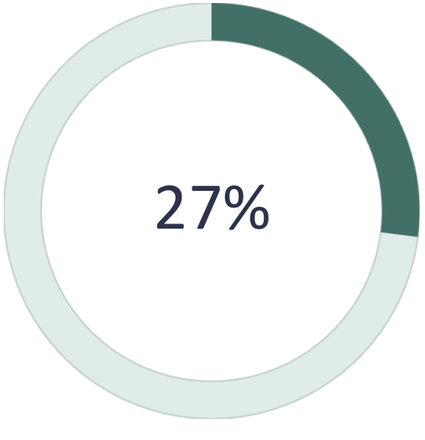

Female

1,250 customers




Non-Subscribers

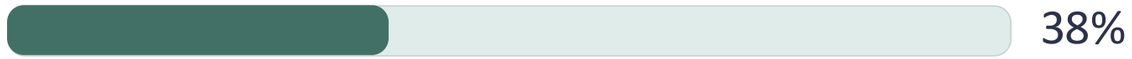
2,850 customers




Subscribers

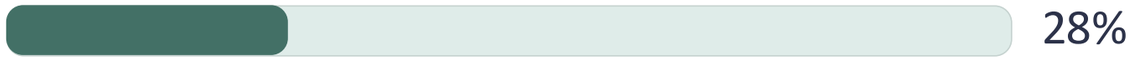
1,050 customers

# Revenue by Age Group



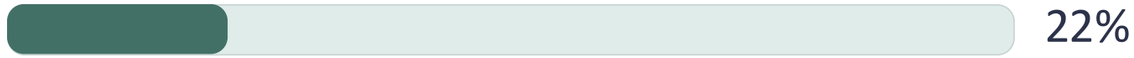

Adult (30-49)

$88,540 revenue



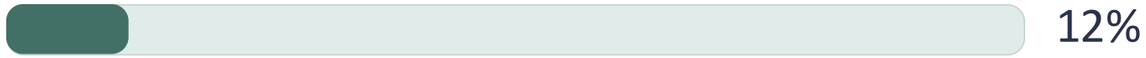

Middle Aged (50-64)

$65,120 revenue




Young Adult (0-29)

$52,340 revenue




Senior (65+)

$27,081 revenue

# Revenue by Category

Clothing

$104K revenue

Accessories

$74K revenue

Footwear

$36K revenue

Outerwear

$19K revenue

# Key Findings

### Adult Segment

30-64 age group drives highest revenue revenue

### Retention Gap

73% non-subscribers represent growth growth opportunity

### Clothing Leader

$104K revenue, outerwear has growth growth potential

# Strategic Recommendations

01

## Expand Subscription Incentives

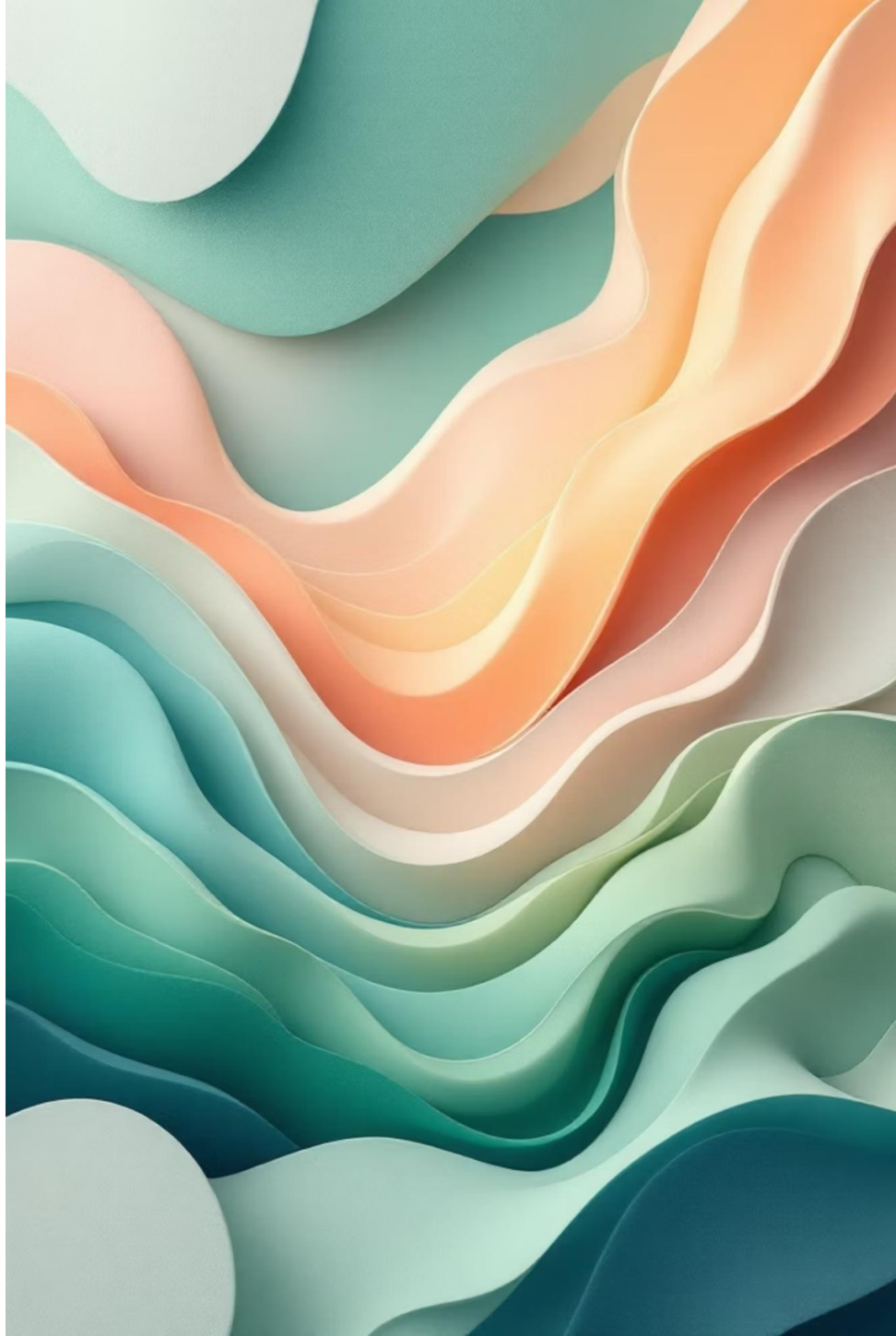
Convert 73% non-subscribers through targeted offers for returning customers

02

## Target 30-64 Age Segment

Focus marketing on highest lifetime lifetime value demographic

03

## Boost Outerwear Sales

Seasonal promotions in late summer for winter items